

University of the Philippines-Diliman
Department of Computer Science

CS 210
Sample Exam 2

11 April 2016
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DIRECTION: Answer all problems clearly and completely specifying all justifications for arguments which are not obvious (to a 5 year old kid). Label all your variables properly using notations adopted in class whenever possible. Good luck!

Notations:

- ϵ — denotes a neutral object
- $\mathcal{E}$ — denotes the class that contains only a neutral object
- $\mathcal{Z}_a$ — denotes the class that contains only an atom a

1. Make an asymptotic expansion to within the given O -approximation of the following:
 - a) Expand $\sin(x) \cdot \ln(x^2 + 1)$ as $x \rightarrow 0$ to within $O(x^6)$.
 - b) Expand $\frac{N+2}{N^2-1}$ as $N \rightarrow \infty$ to within $O(N^{-4})$.
 - c) Calculate an asymptotic expansion for $\binom{3N}{N}/(N+1)$.
 - d) Give an asymptotic estimate for $\sum_{k=1}^N \frac{1}{k^2 H_k}$.
2. Do as you are told in the following items:
 - a) Estimate the value of $e^{0.1} + \cos(0.1) - \ln(0.9)$ to within 10^{-4} , without using a calculator.
 - b) Show that
$$\frac{1}{9801} = 0.00010203040506070809101112 \dots 47484950$$
to within 10^{-100} .
3. Let $\mathcal{B}_{111}$ be the set of all bitstrings containing no three consecutive 1's.
 - a) Write a combinatorial construction for $\mathcal{B}_{111}$.
 - b) If $B_{111}(z)$ is the OGF of $\mathcal{B}_{111}$, what is the corresponding functional equation satisfied by $B_{111}(z)$.
 - c) Find $[z^N]B_{111}(z)$.
4. Let $\mathcal{U}$ be the set of binary trees with size of a tree defined by the total number of nodes (internal plus external).
 - a) Make a combinatorial construction for $\mathcal{U}$.
 - b) Derive an explicit expression for the OGF $U(z)$ of $\mathcal{U}$.
5. Consider the notation $(p_0 p_1 p_2 \dots p_n)$ for the cycle $p_0 \rightarrow p_1 \rightarrow_2 \rightarrow \dots \rightarrow p_n \rightarrow p_0$. Compute the following: $\{(1\ 3\ 2) + (1\ 2)\} \star (1\ 3\ 2)$ where '+' is the disjoint union and ' $\star$ ' is the labelled product.
6. Derive an EGF for the number of permutations whose cycles are all of odd length.

End of Examination: 50 points